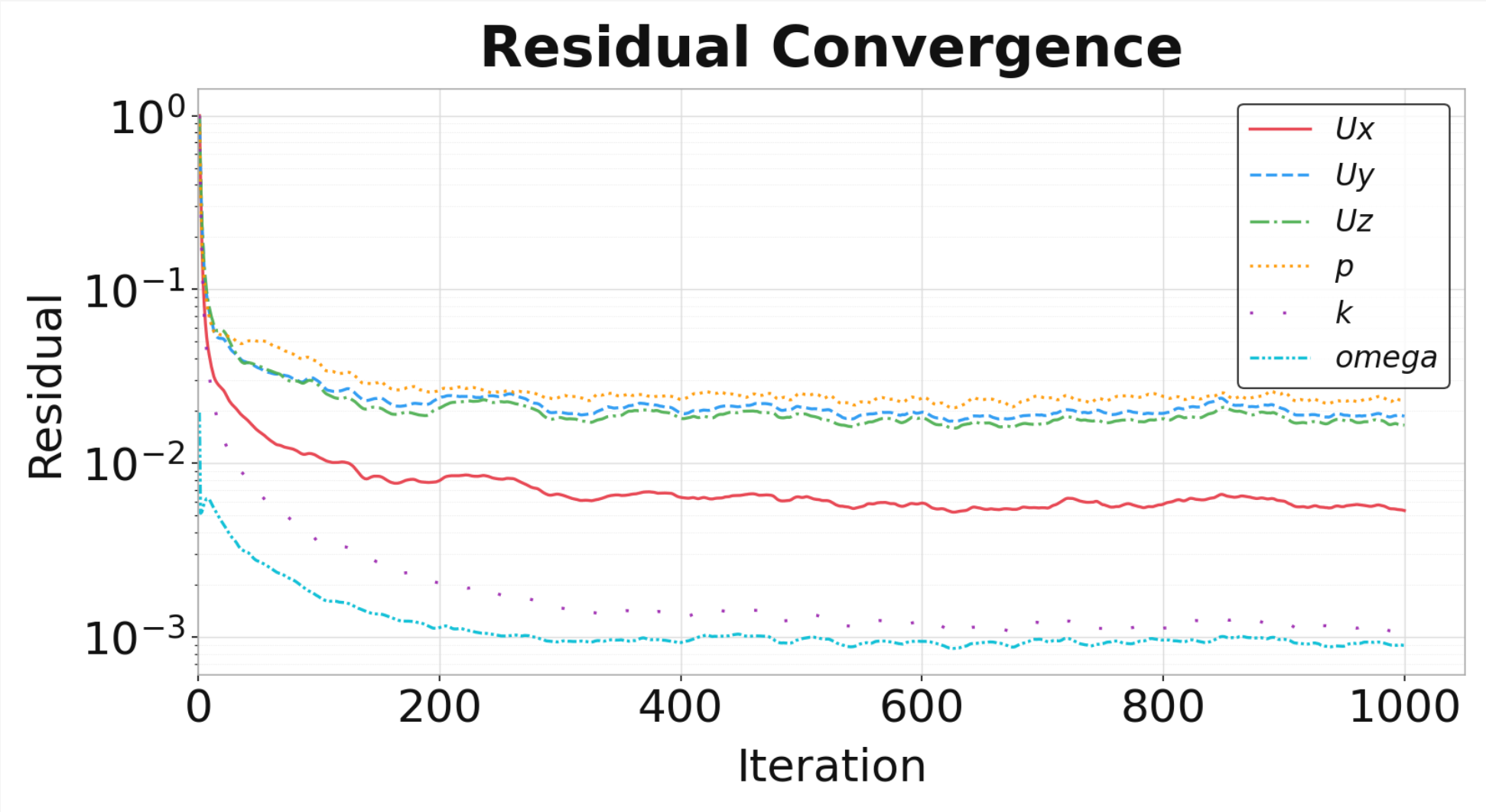
CFD Post-Processing Report

F1 V2

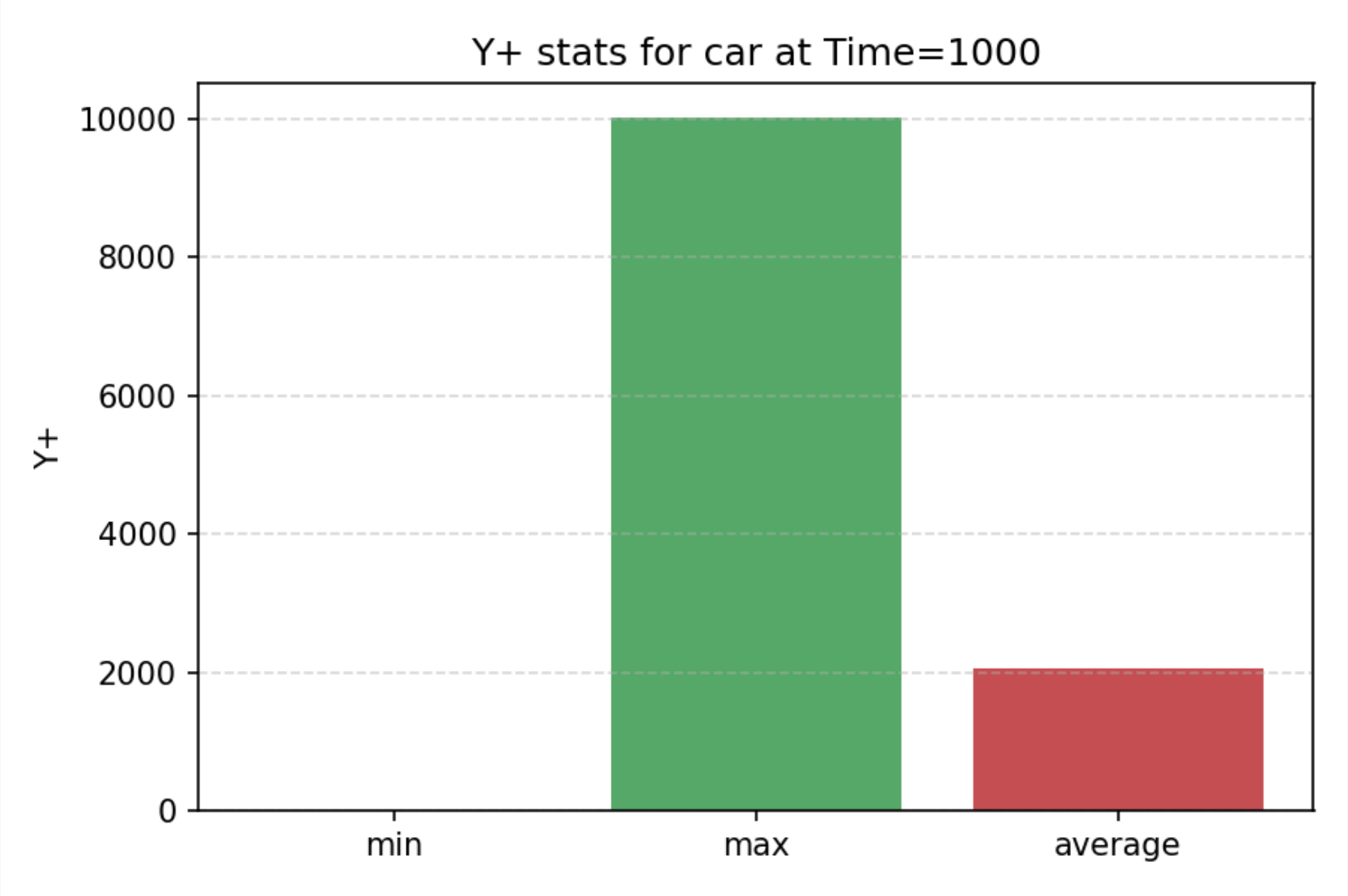
Residuals, Y+ and flow visualisation

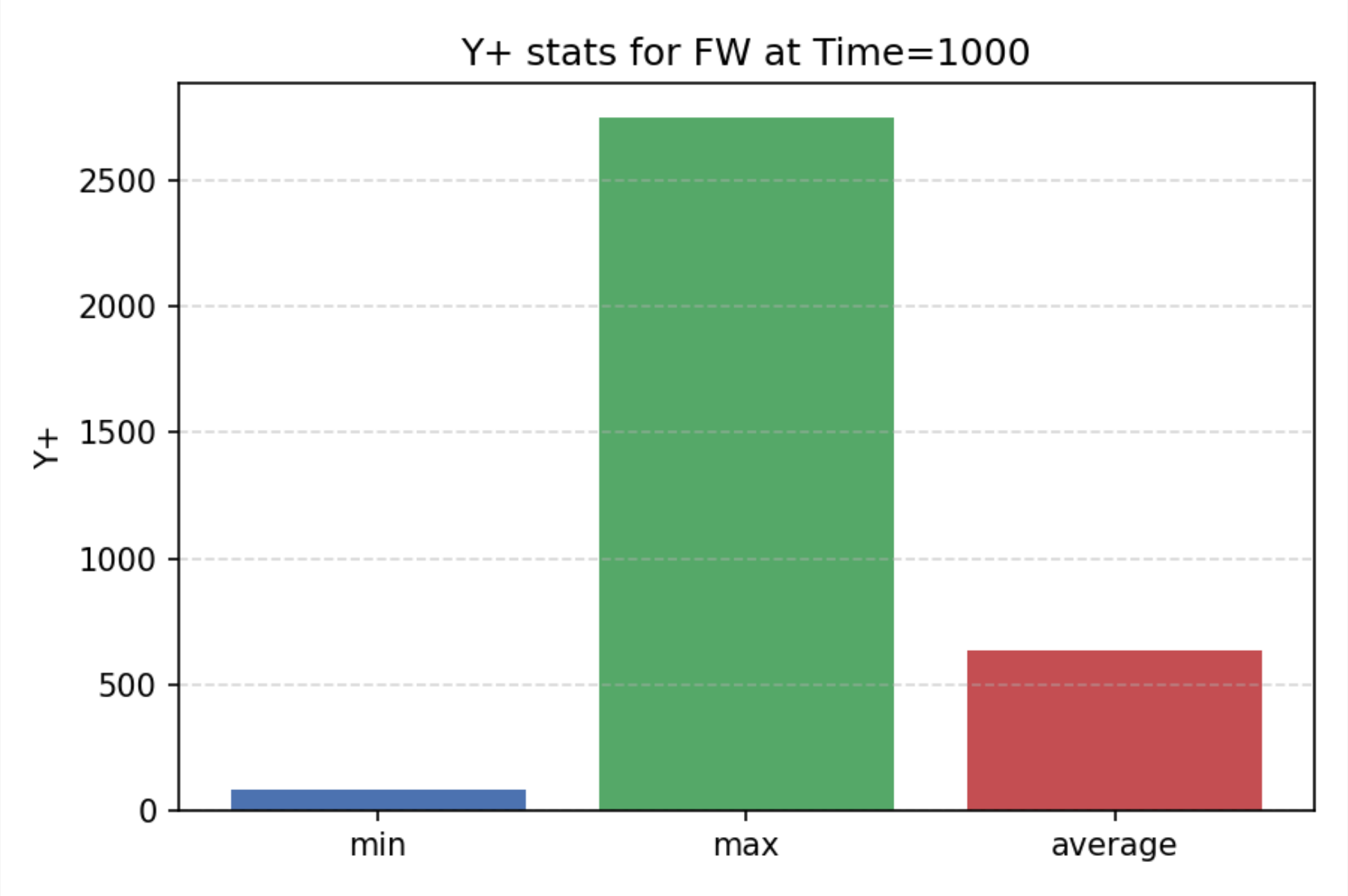
M. Toffoli | 2026-04-15 18:02

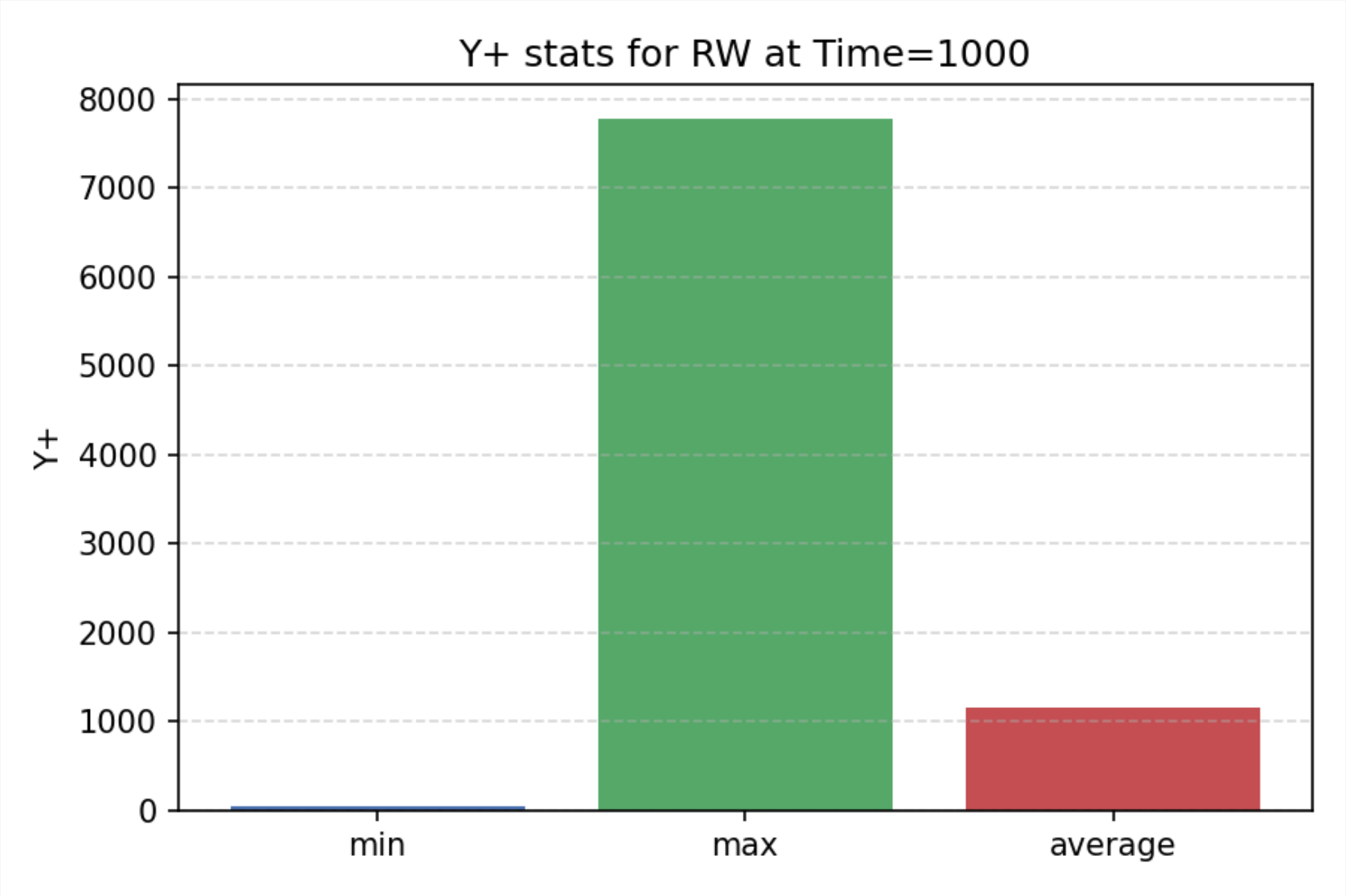
Residuals

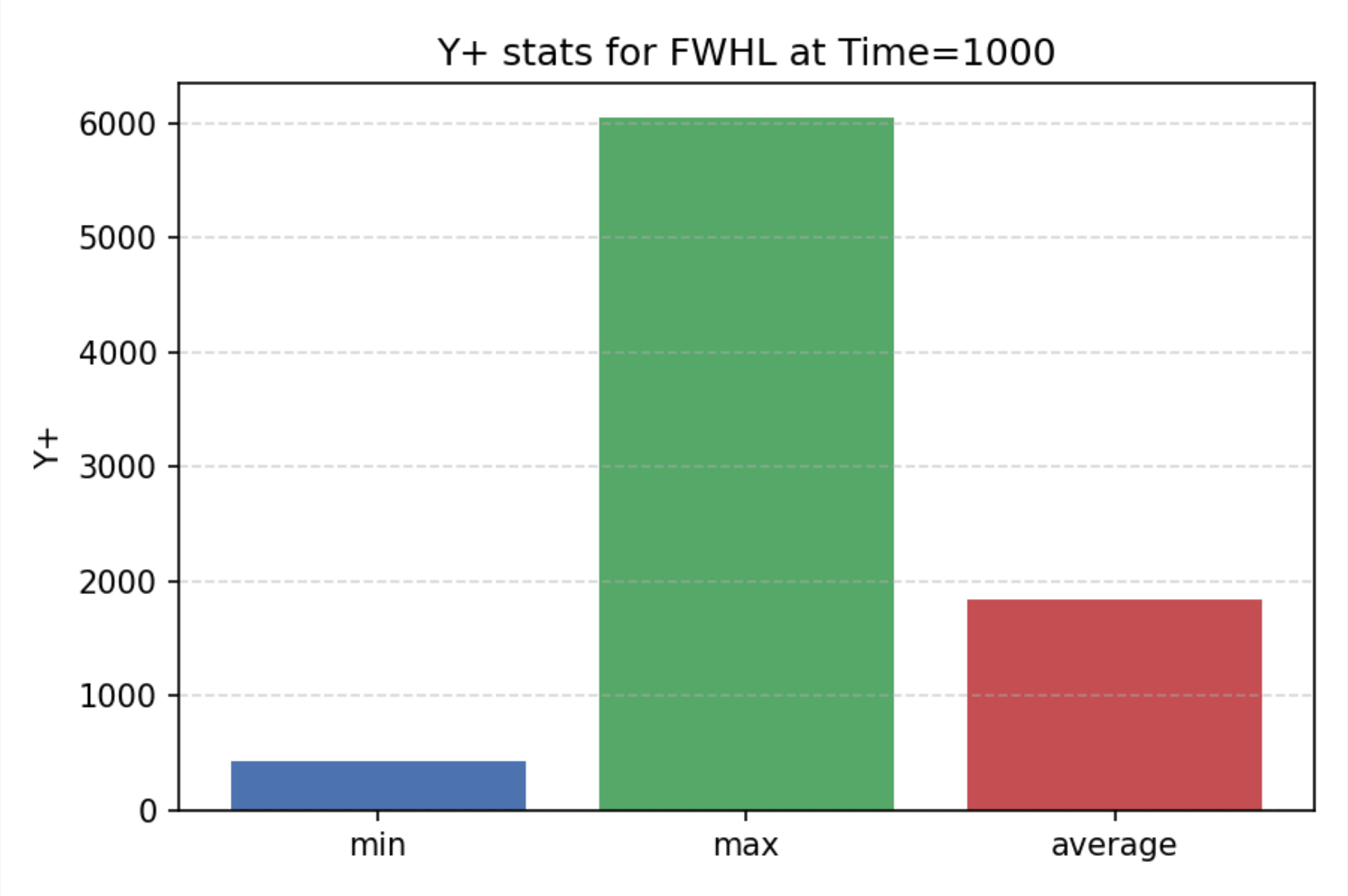

Residual Convergence
Residual
Iteration
10^0
10^{-1}
10^{-2}
10^{-3}
0
200
400
600
800
1000
Ux
Uy
Uz
p
k
omega

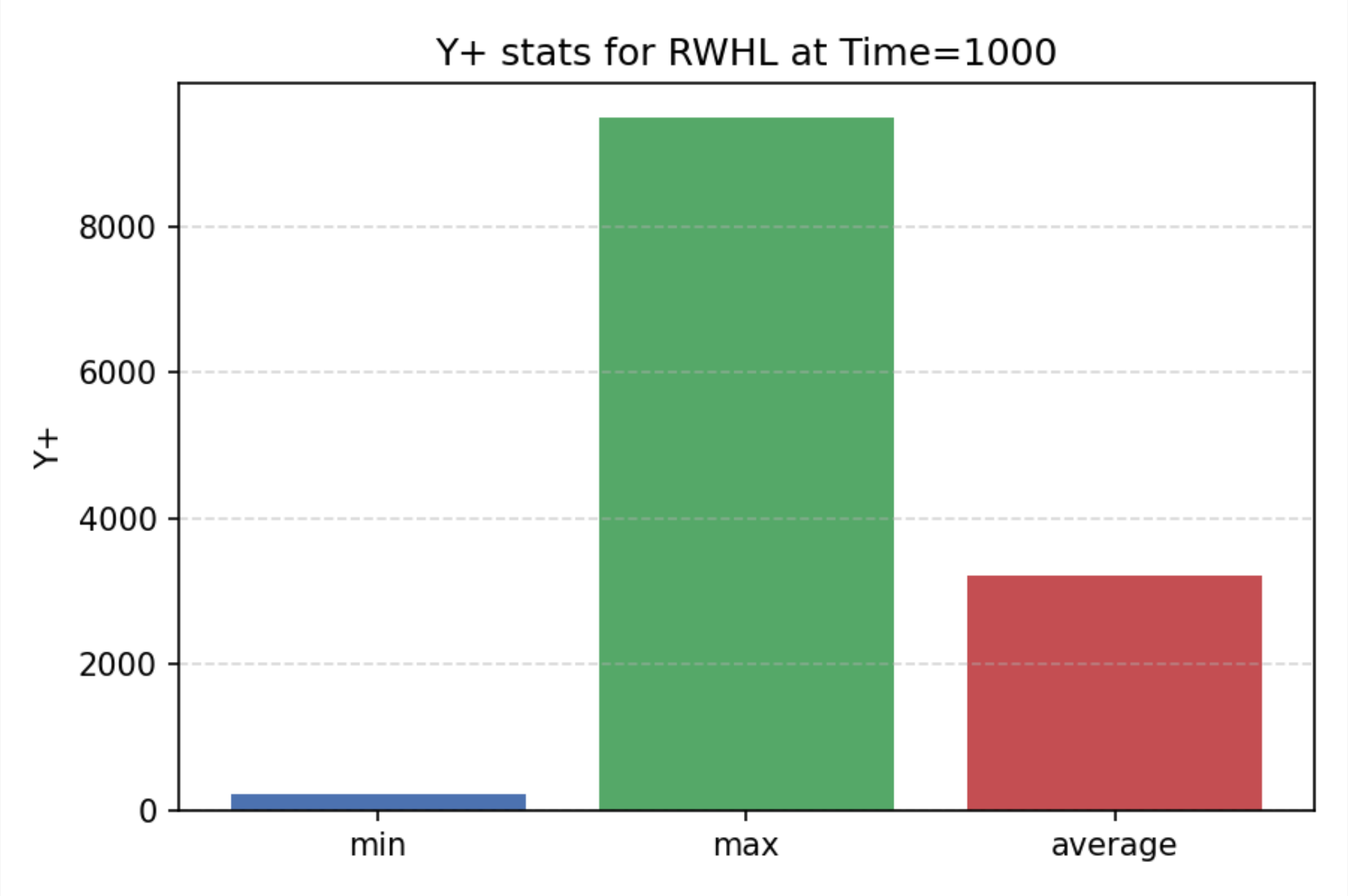
Wall Y+


Y+ stats for car at Time=1000
Y+
10000
8000
6000
4000
2000
0
min
max
average

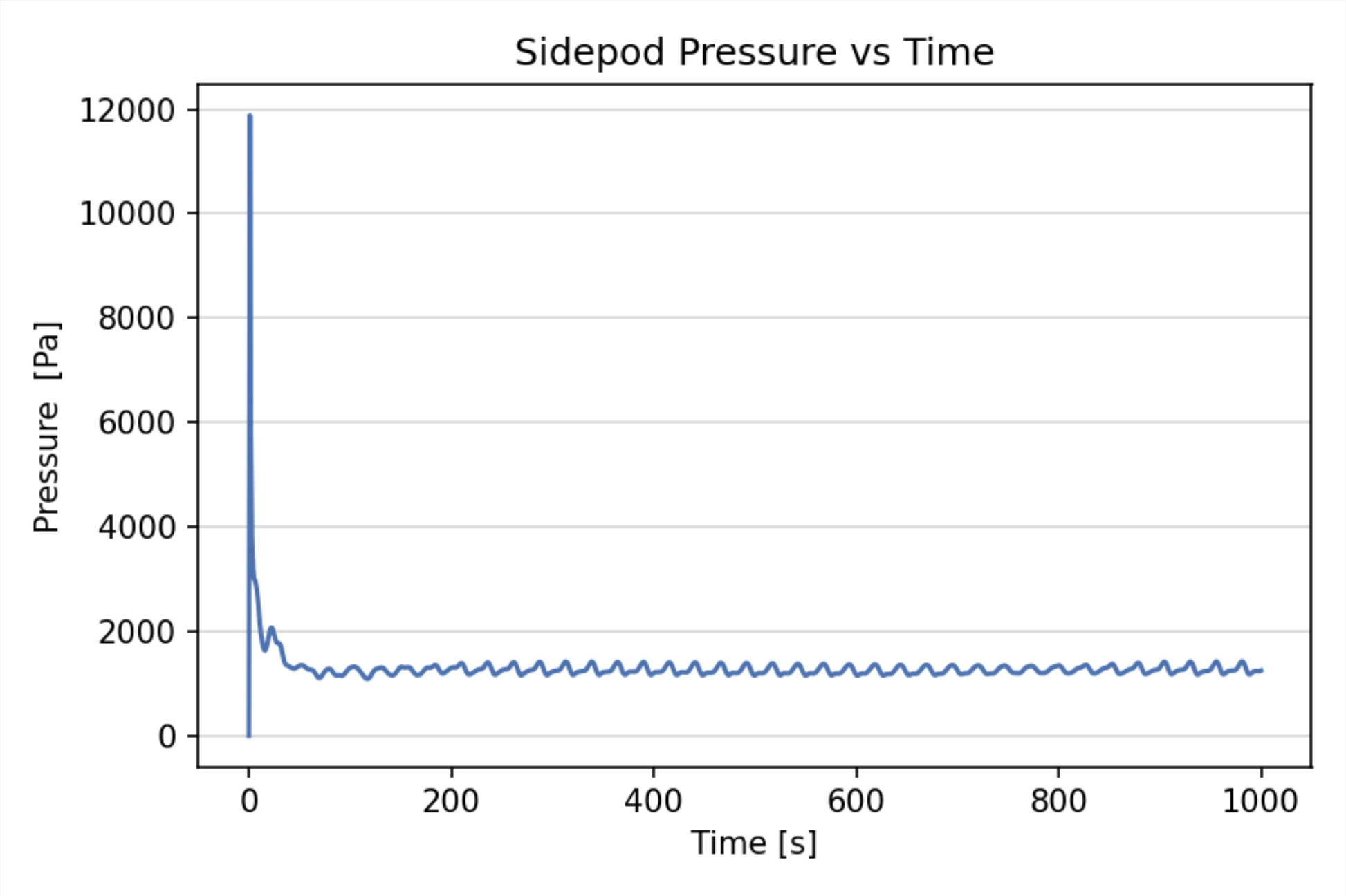

Y+ stats for FW at Time=1000
Y+
2500
2000
1500
1000
500
0
min
max
average


Y+ stats for RW at Time=1000
Y+
8000
7000
6000
5000
4000
3000
2000
1000
0
min
max
average

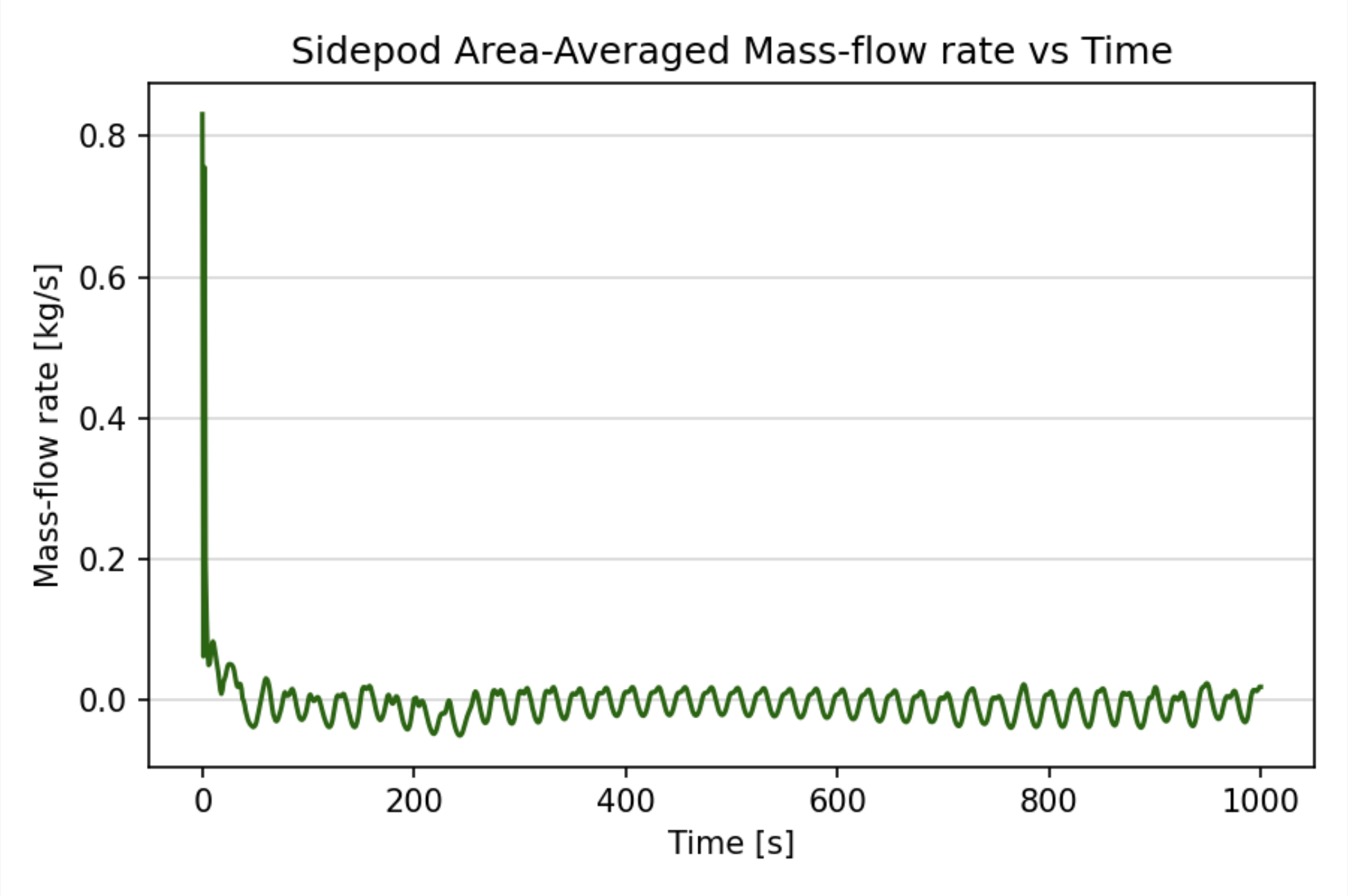

Y+ stats for FWHL at Time=1000
Y+
6000
5000
4000
3000
2000
1000
0
min
max
average


Y+ stats for RWHL at Time=1000
Y+
0
2000
4000
6000
8000
min
max
average

Side Pod Pressure


Sidepod Pressure vs Time
Pressure [Pa]
12000
10000
8000
6000
4000
2000
0
0
200
400
600
800
1000
Time [s]

Side Pod Mass Flow


Sidepod Area-Averaged Mass-flow rate vs Time
Mass-flow rate [kg/s]
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
Time [s]

Expert Mode — Simulation Setup

Solver Setup (controlDict)

Parameter	Value
deltaT	1
endTime	1000
startTime	0
startFrom	startTime
stopAt	endTime
writeInterval	50
writeControl	timeStep
purgeWrite	0
writeFormat	binary
writePrecision	6
timePrecision	6
functionObjects	yPlus, forceCoeffs, wallShearStress, residuals, sidepodPlaneExport, midPlaneCut, …

Numerical Schemes (fvSchemes)

Parameter	Value
ddtSchemes → default	steadyState
gradSchemes → default	Gauss linear
gradSchemes → grad(U)	cellLimited Gauss linear 1
divSchemes → default	none
divSchemes → div(phi,U)	bounded Gauss linearUpwindV grad(U)
divSchemes → div(phi,k)	bounded Gauss upwind
divSchemes → div(phi,omega)	bounded Gauss upwind
divSchemes → div((nuEff*dev2(T(grad(U)))))	Gauss linear
laplacianSchemes → default	Gauss linear corrected
interpolationSchemes → default	linear

snGradSchemes → default	corrected

Solution Control (fvSolution)

Parameter	Value
solver(p)	GAMG tol=1e-7 relTol=0.01
solver(Phi)	? tol= relTol=
solver(U)	smoothSolver tol=1e-8 relTol=0.1
solver(k)	smoothSolver tol=1e-8 relTol=0.1
solver(omega)	smoothSolver tol=1e-8 relTol=0.1
SIMPLE.nNonOrthogonalCorrectors	0
SIMPLE.consistent	yes
potentialFlow.nNonOrthogonalCorrectors	10
relaxation.equations.U	0.9
relaxation.equations.k	0.5
relaxation.equations.omega	0.5

Decomposition (decomposeParDict)

Parameter	Value
numberOfSubdomains	32
hierarchicalCoeffs.n(4 4 2)	
hierarchicalCoeffs.order	xyz

Turbulence Model (momentumTransport)

Parameter	Value
simulationType	RAS
RAS.model	kOmegaSST

RAS.turbulence	on

Physical Properties

Parameter	Value
nu	[0 2 -1 0 0 0 0] 1.5e-05